

# Syllabus - Project 2-1

## Module Adaptive Systems - Artificial Intelligence and Machine Learning (AIML)

2026-2027

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# 1 Project 2-1 Basic Information

Course Code: BCS2720

Title: Project 2-1

ECTS: 6

Programme: BSc Computer Science Adaptive Systems

Study Year: 2

Period: 2.1-2.3

## Contact Information

This information is provided on Canvas.

**PLEASE NOTE:** *This document only serves as an overview. It is subject to change during execution. Any updates will be made directly on the Canvas page. In case of discrepancies, please always refer to the information provided on Canvas as the most up-to-date source.*

# 2 Project coordination

The examiners, coordinator, and tutors for project 2-1 are/will be listed on the page titled *People in this project* on Canvas. General information and course and schedule details can be found on Canvas.

- For content questions: Try to find online or contact experts on this topic. Tutors are not content experts (maybe teachers of classes can help or staff at DACS who research on the topic)
- For group dynamic issues: Contact mediators (they do not reveal anything if you ask them to be confidential. But you can always also contact the tutor and/or coordinator)
- For questions about organization, such as scheduling, deadlines, deliverables, etc. Please contact the coordinator.
- For questions about skill classes: Ask the skill class teacher and put the coordinator in cc.

# 3 Course Description and Intended Learning Outcomes

Students work on a project assignment in small groups of about seven students. The group composition stays the same for the whole project and is announced at the beginning of period 2.1. The students are guided through the project by a fixed tutor (subject to change based on tutor availability). The project assignment is divided into three subtasks (one per period) and is related to the content of the courses from period 2.1 and 2.2. In periods 2.1 and 2.2, the

students work on the project, while also having to attend the courses of these periods. They meet their tutor approximately once every two weeks. In period 2.3, the students work three weeks full-time on the project and meet their tutor every week.

### **3.1 ILOs**

Applying knowledge and understanding: Students will learn to concretize a project assignment and construct and maintain a plan. Furthermore, they will learn formulating, selecting and validating models for a given problem, collect, and interpret experimental data with evaluation metrics. Lastly, they will improve their ability to plan and chair meetings, create minutes for meetings, work in a team such that the workload is balanced, and plan teamwork by setting deadlines and distributing tasks.

Making judgements: Students will be able to compare and criticize results, position them in terms of the literature, diagnose limitations, and formulate a discussion.

Communication: Students will be able to write a scientific paper that: describes the project, explains the methods, summarizes the outcomes, discusses them, and draws conclusions. Students will be able to present and defend their choices in English and coordinate project progress in project meetings

Learning skills: Students will be able to reflect on the progress of the project and study relevant literature to solve the problem at hand.

#### **3.1.1 Knowledge and understanding**

- Understand and interpret the project description

#### **3.1.2 Applying knowledge and understanding**

- Concretize project assignment
- Construct and maintain a planning
- Formulate, select, implement, and validate models for the game chosen
- Collect and interpret experimental data with evaluation metrics
- Identify project goals, deliverables, and constraints.
- Plan and structure meetings by dividing roles, deciding on the meeting content and length
- Work with others by assigning and completing tasks

### **3.1.3 Making judgement**

- Compare and criticize results, position them in terms of the literature, diagnose limitations, and formulate a discussion
- Study relevant literature to solve the problem at hand

### **3.1.4 Communication**

- Give a clear and well-constructed presentation, including a demonstration of the product, and with appropriate use of illustrations and/or videos.
- Offer and respond to questions and constructive criticism of presentations.
- Write a project report according to the structure of an academic article.
- Submit arguments in exact sciences, with appropriate use of formulae and figures.
- Cite published sources in the project report according to the academic guidelines.
- Structurally inform stakeholders on project progress.
- Effectively communicate with project group members about task division, planning and project deadlines.
- Effectively communicate with group members by listening to others' ideas; be contactable and include others in the discussion.
- Cooperate in a group to reach a consensus view.
- Communicate in the English language

### **3.1.5 Learning skills**

- Reflect on the progress of the project

## **4 Overview of the subjects/topics**

See project manual descriptions.

## **5 Study load (i.e. course workload)**

This project is part of the module; the module is worth 10 ECTS in total, of which this project is worth 6 ECTS points, which equates to (individual) 168 hours of study in periods 1-3. In periods 1 and 2, you are expected to work about 4 hours a week on the project, and in period 3, you are expected to work full-time, i.e., 8 hours a day on the project.

## 6 Project Materials

- Project opening slides
- Project manual
- Syllabus
- Rules and Regulations: read them carefully. Any emails that can be answered by the rules and regulations document will be ignored

## 7 Project timeline

This project is part of the elective module AI and Machine Learning - the module consists of the Machine Learning course, skill classes, and this project on Adaptive Systems.

The project assignments in periods 1 and 2 are preparatory for the concluding part of project 2-1, which takes place in period 3. At the beginning of period 1, the project starts in week 1 with a project opening. In periods 2 and 3, the group will start working on the project immediately without having a separate project opening. While in periods 1 and 2, the group will work part-time on the project (simultaneously with your courses); in period 3, the group will work full-time on the project. At the end of period 1, the group hands in a detailed project plan, that includes an outline of the work accomplished so far. By the end of period 2, the Midway Evaluation takes place: the group presents their progress with a presentation and demo, and receives feedback from the examiners. At the end, the project is assessed by a final report, and a Project Defense, in which the students presents their work and product, and the examiners ask (individual) questions (non-public).

## 8 Project meetings

A project meeting with the tutor aims to continuously track the status of the project. The planning is checked, and agreements and appointments are made and checked. In case of deviations from the planning, an analysis of the situation is made in order to trace the causes and remedy the situation. Project meetings are scheduled on a fixed date and time. Usually, these time slots contain 4 hours (in exceptional cases, 2 hours may be scheduled). The students are requested to work these 4 hours on the project. The tutor visits the groups one by one in a pre-defined time slot of 2 hours and 10 minutes. Attendance is tracked.

Here are the specific steps that students and tutors are supposed to follow:

- Before the tutor joins the scheduled project meeting: Project Group fills out "Tasks completed before the PM" and "Time spent on completed tasks" for each student (updating the group on individual progress)

- The tutor will join the meeting sometime during the scheduled project meeting and discuss with students what they have done individually, also in alignment with the issue board (Please note: Be ready to discuss with the tutor as soon as they come into the room. One group member projects the file on the screen. EVERYONE joins the discussion.)
- Tutor, together with the students, flags the two rows "Tasks completed before the PM" and "Time spent on completed tasks" per student according to:
- If needed, tutor adds comments e.g. if needed time sounds too long, or tutor has doubts about honesty, or student is absent)
- Tutor discusses open questions/project management/etc.
- Tutor leaves
- Group continues working on their tasks until the end of the scheduled project meeting.
- Last fifteen to thirty minutes of scheduled project meeting: Group fills out the "Tasks to be completed before next PM" and the "Estimated time" at the end of the scheduled project meeting. Students with a yellow/red flag reflect on their flag by: 1) Description: Provide a clear, factual account of the situation without analysis or judgment. Example: Our team had to train and evaluate a neural network on a text dataset. I was assigned hyperparameter tuning but submitted incomplete code, causing a delay. 2) Analysis: Why did it happen? Example: I lacked confidence in the library to use and avoided working on it until it was too late; 3) Action: Outline specific steps you will take to improve in the future. Example: I'll test code in smaller chunks and commit regularly so teammates can help if needed.
- Before the end of the project meeting, group uploads the files to Canvas

## 8.1 GitHub as project management tool

The students will use GitHub to keep track of their tasks and their status. The tutor and examiners will have access to the groups' project repository to be able to monitor the progress as well. Each task will be assigned to members of the group. The students will use the Issues tab to keep track of their tasks and their status. Together with the tutor, the groups' project progress is monitored and reflected in each project meeting.

# 9 Grades

## 9.1 Examination moments

- Period 1: Graded project plan

- Period 2: Midway Evaluation (formative, not graded)
- Period 3: Final presentation and project defence

## 9.2 Project grade

Students receive a group grade that is calculated in the following way:

$$\text{GroupGrade} = 0.15 \cdot \text{ProjectPlan} + 0.1 \cdot \text{ProjectManagement} + 0.4 \cdot \text{Product\&Defence} + 0.35 \cdot \text{Report}$$

and

$$\text{FinalGrade} = 0.5 \cdot \text{CourseGrade} + 0.4 \cdot (\text{GroupGrade OR IndividualGrade}) + \text{SkillClassGrade}$$

where the examiners assess the project plan, product, and report, and the tutors provide input for the project management.

The skill class grade is calculated as follows: *skillClassGrade* = 1, if the student passed all skill classes, = 0.5, if the student passed all but one skill classes, = 0 if the student passed all but two skill classes. (NG for the project is given if a student failed more than two skill classes). If you fail 3 skill classes or more without a valid reason (to be assessed by the Board of Examiners), you will receive an NG for the project.

Note that the Board of Examiners only looks into your evidence if this grade reduction leads to NG or a failing final grade (not for skill classes). Do **not** contact the Board of Examiners unless explicitly instructed to do so by the project coordinator!

Students not complying with attendance and participation requirements during the project meetings or examination moments may not be allowed to attend examination moments and may receive an NG.

Furthermore, examiners can give an individual grade based on the contribution of the students. In the case that the contribution of a student deviates extensively from the overall performance of the group, this student can receive an individual grade. Such a deviation can result in an increased or decreased grade or in an NG.

Feedback on assessment will be provided by using rubrics in Canvas assignments.

More information on the assessment can be found in the rules and regulations and assessment forms uploaded to Canvas. It is highly recommended to read them carefully. To pass the project, you need a final grade higher or equal to 6 (note: a 5.5 is not rounded to 6 and is a failing grade).

### 9.3 Attendance and participation

On-campus attendance and participation in the project meetings and project skills trainings, are mandatory. Missing a meeting/training in this article means failure to be present during part of or the complete meeting/training, inadequate participation, or inadequately completing the assignments. For project meetings, the project tutor or project coordinator decides whether a student missed it or not, and for the skill trainings, the lecturer or project coordinator decides.

All project meetings are mandatory. Project meetings include, but are not limited to, the project opening meeting, progress meetings with examiners, and any additional meetings announced by the coordinator. Students may miss up to three project meetings without consequences. Nevertheless, students are responsible for informing the relevant stakeholders of their absence and for catching up on any missed work.

If a student is absent from more than three project meetings, this results in an NG (no grade); appropriate supporting evidence must be provided to the BoE (only when the coordinator instructs the students to do so) for all missed project meetings to get an exemption from an NG. These hardship cases (for definitions, see document Rules and Regulations) are considered highly individual and are expected to be very rare. Students who encounter such circumstances should contact the coordinator in the first instance; the BoE will make the final decision where required. Any repair procedure will include an oral examination.

Participation in project skill trainings is taken into account in the final grading. If a student has missed three or more skills trainings, the student will not be able to participate in the final examination of the project, meaning that an NG is given for the project. This NG will also be awarded if, after the final examination, it turns out that an ineligible student still participated. The same holds when a student misses too many project meetings.

Failure to participate in the Midway Evaluation leads to an automatic grade reduction. Failing to participate in the final examination leads to an NG.

The arrangements that offer the possibility to miss project meetings and skill classes should not be taken lightly and only be used in a case of clear force majeure.

### 9.4 Resit

Students who scored  $< 6.0$  receive an additional assignment or required corrections/rework of the original assignment that needs to be completed inside of 10 working days. The exact assignment is dependent on the performance of the group in the original project. The examiners provide a clear definition of the new, additional, or repeated assignment within 4 working days after the original grade.

If you receive NG for the project or skill classes, you are not entitled to a repair assignment.

## **10 Assessment Components**

### **10.1 Project Plan**

At the end of period 2.1, you submit a written project plan, in which you write a detailed plan for the steps to take over the duration of the project. The plan should include a rough schedule, and an assessment of risks and how to work around them. The maximum number of pages is ten, though a lower number of pages is also fine.

### **10.2 GitHub repository**

The group also creates one publicly visible GitHub repository, with relevant documentation, to contain your work. You continue iterating on this page throughout the entire project.

The examiners will pay attention to code quality, but also other aspects of the repository (e.g., documentation, proper use of git commit messages and other features).

### **10.3 Midway Evaluation**

The Midway Evaluation is a formative assessment, in which the examiners check the progress of the group and provide feedback and advice. It takes place halfway through the project, in period 2. It is an in-person examination moment that lasts 20 minutes. By the end of period 2.2, you give a demonstration of your work and explain it in an in-person presentation (at least 5 minutes should be reserved for Q&A, and preferably more).

The group should bring an additional laptop with code ready to be checked and run.

### **10.4 Report**

You submit a final report by the end of the project. We uploaded writing guidelines to Canvas. Please read them carefully. In project 2-1, you have to write your report in LaTeX. The page limit is 10 pages (excluding any appendices and references) in a double-column layout.

The report should at least include an explanation of the problem setting, research questions, and discussions of the implemented (and other considered) approaches, experiments and results, a conclusion with answers to the research questions, and lessons learned.

## **10.5 Project defense**

The Product and Report Examination takes 30min for each group. You prepare a short presentation of your experimental work and a demo. This is the opportunity for the examiners to ask (individual) questions about the product and report to assess the understanding of the students (at least 15min). The examiners do not need to inform the students if they plan to ask individual questions.

## **11 Submissions**

### **11.1 Group Portfolio**

Every group is expected to create a Group Portfolio based on the provided template to give an overview of what every group member did during the project, in the project plan, at the midway evaluation, and at the final submission. It is highly recommended to start keeping track of this at the beginning of the project. It is uploaded along with the other assessment components to Canvas. If you do not agree on an overview, let your tutor and the coordinator know within a working day after submission.

### **11.2 Overview of who did what**

Every group is expected to create an overview of what every group member did during the project, in the project plan, at the midway evaluation, and at the final submission. It is highly recommended to start keeping track of this at the beginning of the project. It is uploaded along with the other assessment components to Canvas. If you do not agree with an overview, let your tutor and the coordinator know within a working day after submission.

### **11.3 Project Plan**

At the end of period 2.1, you submit a written project plan, in which you write a detailed plan for the steps to take over the duration of the project. The plan should include a rough schedule and an assessment of risks and how to work around them. The maximum number of pages is ten, though a lower number of pages is also fine.

The project plan should not be longer than 10 pages, including the Gantt chart, and it counts for 15% of the project grade.

### **11.4 Planning**

Every group is expected to make a plan at the beginning of phase two and phase three for the complete phase. The planning should give an overview of when the groups expect to achieve which milestones and who is going to work on

which tasks. It is uploaded to Canvas in the corresponding phase (see Canvas for concrete deadlines).

## **12 Schedules**

The concrete schedules for the project meetings and the presentations will be uploaded to Canvas.

## **13 Rules and Regulations**

The rules and regulations can be found on Canvas. They include, e.g., the consequences for missing mandatory events (project meetings, skill classes, assessment moments). Please read them carefully. Any contradiction to the rules and regulations and/or the EER should be ignored. The RR and the EER are the official documents that count.